
Curriculum Vitae of Ilyas Fatkhullin

Address: Department of Computer Science, Andreasstrasse 5, 8050 Zurich, Switzerland
Website: fatkhullin.github.io Email: ilyas.fatkhullin@ai.ethz.ch, fatkhullin.ilyas@gmail.com

Current position: Applied Scientist Intern at Amazon, Forecasting & Optimization Team, Luxembourg.

Education

ETH Zurich, Switzerland

Ph.D. in Computer Science, ETH AI Center Fellow 12/2021 – 06/2026
Advisor: Prof. Niao He (expected)

Research visit: **Georgia Institute of Technology**, USA (with Prof. Guanghai Lan), 03–09/2025

Dissertation Topic: From Hidden Convexity to Stochastic Dynamics in Reinforcement Learning

Technical University of Munich, Germany

M.Sc. in Mathematics Grade: 1.1/1.0 09/2020 – 11/2021

Advisor: Prof. Peter Richtárik

Formal Reviewer: Prof. Michael Ulbrich

Thesis Topic: Error Compensation Method for Compressed Distributed Training

Moscow Institute of Physics and Technology, Russia

B.Sc. in Mathematics and Informatics Grade: 4.9/5.0 09/2016 – 08/2020

Advisor: Prof. Boris Polyak

Thesis Topic: Optimization Landscape of Linear Quadratic Regulator Problem with Output Feedback

Publications

The most up-to-date list of publications is available on my [Google Scholar](#) profile.

Refereed Journal Papers

- [1] I. Fatkhullin, N. He, Y. Hu. *Stochastic Optimization under Hidden Convexity*. [SIAM Journal on Optimization](#), 2025.
- [2] J. Wu, A. Barakat, I. Fatkhullin, N. He. *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity and Last-Iterate Convergence*. [SIAM Journal on Control and Optimization](#), 2025 (preliminary version at [CDC](#), 2023).
- [3] I. Fatkhullin, I. Sokolov, E. Gorbunov, Z. Li, P. Richtárik. *EF21 with Bells & Whistles: Six Algorithmic Extensions of Modern Error Feedback*. [Journal of Machine Learning Research](#), 2025.
- [4] I. Fatkhullin, B. Polyak. *Optimizing Static Linear Feedback: Gradient Method*. [SIAM Journal on Control and Optimization](#), 2021.
 - *Cited over 100 times*.
- [5] B. Polyak., I. Fatkhullin. *Use of Projective Coordinate Descent in the Fekete Problem*. [Computational Mathematics and Mathematical Physics](#), 2020.

Refereed Conference Papers (ML)

- [1] A. Sadiev, P. Richtárik, I. Fatkhullin. *Second-order Optimization under Heavy-Tailed Noise: Hessian Clipping and Sample Complexity Limits*. [NeurIPS](#), 2025.
- [2] F. Sun, I. Fatkhullin, N. He. *Natural Gradient VI: Guarantees for Non-Conjugate Models*. [NeurIPS](#), 2025.
- [3] R. Islamov, Y. As, I. Fatkhullin. *Safe-EF: Error Feedback for Nonsmooth Constrained Optimization*. [ICML](#), 2025.

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- [4] F. Hübler*, I. Fatkhullin*, N. He. *From Gradient Clipping to Normalization for Heavy-Tailed SGD*. [AISTATS](#), 2025. *Equal contribution.
 - [5] I. Fatkhullin, N. He. *Taming Nonconvex Stochastic Mirror Descent with General Bregman Divergence*. [AISTATS](#), 2024.
 - [6] I. Fatkhullin, A. Tyurin, P. Richtárik. *Momentum Provably Improves Error Feedback!* [NeurIPS](#), 2023.
 - [7] J. Yang, X. Li, I. Fatkhullin, N. He. *Two Sides of One Coin: the Limits of Untuned SGD and the Power of Adaptive Methods*. [NeurIPS](#), 2023.
 - [8] A. Barakat, I. Fatkhullin, N. He. *Reinforcement Learning with General Utilities: Simpler Variance Reduction and Large State-Action Space*. [ICML](#), 2023.
 - [9] I. Fatkhullin, A. Barakat, A. Kireeva, N. He. *Stochastic Policy Gradient Methods: Improved Sample Complexity for Fisher-non-degenerate Policies*. [ICML](#), 2023.
 - [10] P. Richtárik, I. Sokolov, I. Fatkhullin, E. Gasanov, Z. Li, E. Gorbunov. *3PC: Three Point Compressors for Communication-Efficient Distributed Training and a Better Theory for Lazy Aggregation*. [ICML](#), 2022.
 - [Spotlight Presentation at ICML Main Track](#) (top 5% submissions).
 - [11] I. Fatkhullin*, J. Etesami*, N. He, N. Kiyavash. *Sharp Analysis of Stochastic Optimization under Global Kurdyka-Lojasiewicz Inequality*. [NeurIPS](#), 2022. *Equal contribution.
 - [12] P. Richtárik, I. Sokolov, I. Fatkhullin. *EF21: A New, Simpler, Theoretically Better, and Practically Faster Error Feedback*. [NeurIPS](#), 2021.
 - [Oral Presentation at NeurIPS Main Track](#) (top 1% submissions).
 - [Cited over 200 times](#).

Preprints / Under Review

- [1] I. Fatkhullin, F. Hübler, G. Lan. *Can SGD Handle Heavy-Tailed Noise?*, 2025. Under review in SIAM Journal on Optimization.
 - NeurIPS Workshop “Optimization for Machine Learning”.
 - [Oral Presentation at NeurIPS OPT 2025 Workshop](#). Preprint: [arXiv:2508.04860](#).
- [2] I. Fatkhullin, N. He, G. Lan, F. Wolf. *Global Solutions to Non-Convex Functional Constrained Problems with Hidden Convexity*, 2025. Under review in Mathematical Programming Series A.
 - NeurIPS Workshop “Constrained Optimization for Machine Learning”.
 - [Oral Presentation at NeurIPS COML’25 Workshop](#). Preprint: [arXiv:2511.10626](#).

Fellowships and Awards

Oral Presentation Award from OPT 2025 Workshop at NeurIPS	11/2025
Oral Presentation Award from COML 2025 Workshop at NeurIPS	10/2025
TMLR Expert Reviewer	09/2025
Rising Star in AI Award, KAUST	02/2023
ICCOPT 2022 Travel Grant	05/2022
Spotlight Presentation Award at ICML	07/2022
Oral Presentation Award at NeurIPS	12/2021
ETH AI Center Doctoral Fellowship	02/2021
German Academic Exchange Service (DAAD) Scholarship for MSc in Germany	04/2020
PreDoc Program Fellowship in Mathematics, Technical University of Munich	03/2020

Teaching

Lecturer, “Data Analysis and Machine Learning”, Ashesi University, Ghana
Part of the [Ashesi–ETH Master in Mechatronic Engineering](#).

05/2025

- Independently designed and delivered a 3-week intensive graduate-level course, including syllabus, lectures, coding labs, assignments, a final project, and exam.
- Student evaluations (26/34 response rate): overall instructor rating 4.45/5.00.
- Received particularly strong scores in preparation of lessons (4.54), encouraging participation (4.54), accessibility and approachability (4.58), and use of instructional technology (4.50).

Teaching Assistant, “[Optimization for Data Science](#)”, ETH Zurich

Spring 2023, 2024

- Led weekly tutorials for 100+ students in a core M.Sc.-level optimization course.
- Co-designed homework assignments and contributed exam questions for midterm and final assessments.
- Held office hours and provided individualized support for student project groups.

Instructor, Seminar “Advanced Topics in Machine Learning”, ETH Zurich

Fall 2023, 2024

- Guided research-paper discussions, coached students on presentations and scientific writing.
- Evaluated student presentations and advised on short research exploration projects.

Supervision and Mentoring

Since 2022, I have mentored nine Master’s, semester, and research-project students as a primary supervisor. These collaborations have led to publications in prestigious venues (NeurIPS, ICML, AISTATS, SIAM Journal), and mentees have progressed to Ph.D. programs at ETH Zurich, Caltech, Max Planck Institute.

Ph.D. Supervision

- Rustem Islamov (University of Basel) — Ph.D. project.
- Yarden As (ETH Zurich) — Ph.D. project.

Master’s Theses

- Harish Rajagopal (ETH Zurich, 2022) — subsequently Software Developer, PQFORCE (Switzerland).
- Jiduan Wu (ETH Zurich, 2023) — subsequently Ph.D. student, Max Planck ETH CLS.
- Florian Wolf (TU Darmstadt, 2025) — subsequently Ph.D. student, Caltech.
- Chung-En Tsai (ETH Zurich, current) — Master’s Thesis.

Research Assistants and Interns

- Florian Hübler (ETH Zurich, 2024) — subsequently Ph.D. student, ETH Zurich.

Semester Projects

- Fangyuan Sun (ETH Zurich, 2025) — continued to Master’s Thesis under my supervision at ETH Zurich.
- Lucas Whitfield (ETH Zurich, current).

Research Internships and Short-term Visits

Amazon, Luxembourg

10/2025-02/2026

Applied Scientist Intern in Forecasting & Optimization Team

(ongoing)

Worked on large-scale capacity management problems of Amazon’s delivery network.

Georgia Institute of Technology

H. Milton Stewart School of Industrial and Systems Engineering, Atlanta, USA

03–09/2025

with Prof. Guanghui (George) Lan

Studied properties of stochastic gradient descent under infinite variance.

King Abdullah University of Science and Technology, Saudi Arabia (remote)

03–09/2021

with Prof. Peter Richtárik

Worked on federated learning algorithms with momentum and quantization.

École Polytechnique Fédérale de Lausanne, Switzerland (remote)

06–10/2020

with Prof. Sebastian Stich and Prof. Martin Jaggi

Studied accelerated methods for convex optimization.

German Electron Synchrotron (DESY), Hamburg; with Prof. Judith Katzy

07–09/2019

Applied adversarial learning methods to the detection of events in high-energy physics.

Helmholtz-Zentrum Berlin, Germany; with Prof. Ji Li

07–08/2018

Developed numerical solvers for physical simulations in materials science.

Professional Activities

Journal Reviewer: Mathematical Programming; Operations Research; SIAM Journal on Optimization; SIAM Journal on Control and Optimization; Journal of Machine Learning Research; IEEE Transactions on Automatic Control; Transactions on Machine Learning Research.

Conference Reviewer: ICML (2022-2024); AISTATS 2022; NeurIPS (2023-2025); ICLR 2024.

Session Organizer at INFORMS Optimization Society Conference

07/2024

Location: Rice University, Houston, USA

Cluster and Session Title: “Optimization in Data Science”; “Recent Advances in Min-Max Optimization”

Session Organizer at International Conference on Continuous Optimization

07/2025

Location: University of Southern California, Los Angeles, USA

Cluster and Session Title: “Optimization for Data Science”; “Adaptive Methods in Optimization”

Session Organizer at INFORMS Optimization Society Conference

03/2026

Location: Atlanta, USA

Session Title: “Methods for Multi-stage Decision Making”

Session Organizer at SIAM Optimization Conference

06/2026

Location: Edinburgh, United Kingdom

Session Title: “Optimization under Data-driven Constraints”

Selected Talks

I. Fatkhullin. *Natural Gradient Variational Inference: Guarantees for Non-conjugate Models*.

11/2025

[SlidesLive](#) [NeurIPS 2025](#), San Diego, USA

I. Fatkhullin. *Can SGD Handle Heavy-Tailed Noise?*

08/2025

H. Milton Stewart School of ISyE, Georgia Institute of Technology, Atlanta, USA

I. Fatkhullin. *Safe Error Feedback with Applications in Humanoid Robot Fleet Training*.

08/2025

International Conference on Stochastic Programming (ICSP 2025), Paris, France

I. Fatkhullin. *Nonconvex Stochastic Mirror Descent with Implications in Machine Learning*.

07/2025

International Conference on Continuous Optimization (ICCOPT 2025), Los Angeles, USA

I. Fatkhullin. *Heavy-Tailed Gradients in AI Models*.

12/2024

ETH AI Center Seminar, Zurich, Switzerland

I. Fatkhullin. *Stochastic Optimization under Hidden Convexity*.

07/2024

European Conference on Advances in Continuous Optimization (EUROPT 2024), Lund, Sweden

I. Fatkhullin. <i>Momentum Provably Improves Error Feedback!</i> SlidesLive NeurIPS 2023 , New Orleans, USA	12/2023
I. Fatkhullin. <i>Stochastic Policy Gradient Methods for Fisher-non-degenerate Policies.</i> SlidesLive ICML 2023 , Honolulu, USA	07/2023
I. Fatkhullin. <i>Policy Gradient Methods in Reinforcement Learning.</i> Rising Star in AI Symposium, KAUST, Saudi Arabia	02/2023
I. Fatkhullin. <i>Stochastic Optimization under Kurdyka-Lojasiewicz Condition.</i> Institute of Machine Learning (IML) Seminar, Zurich, Switzerland	10/2022
I. Fatkhullin. <i>Stochastic Optimization under Kurdyka-Lojasiewicz Condition.</i> ETH AI Center Doctoral Seminar, Zurich, Switzerland	06/2022
I. Fatkhullin. <i>Practical Algorithmic Extensions of Modern Error Feedback.</i> Federated Learning One World (FLOW) Seminar	10/2021

References

Prof. Niao He

Associate Professor, Department of Computer Science
ETH Zurich, Switzerland
Email: niao.he@inf.ethz.ch

Prof. Peter Richtárik

Professor of Computer Science
King Abdullah University of Science and Technology (KAUST), Saudi Arabia
Email: peter.richtarik@kaust.edu.sa

Prof. Guanghui (George) Lan

A. Russell Chandler III Chair and Professor, ISyE
Georgia Institute of Technology, Atlanta, USA
Email: george.lan@isye.gatech.edu

Prof. Yifan Hu

Assistant Professor, Department of Statistics
Rutgers University, New Brunswick, New Jersey, USA
Email: yifan.hu@rutgers.edu

Prof. Sebastian U. Stich

Tenured Faculty, CISPA Helmholtz Center for Information Security
Saarbrücken, Germany
Email: stich@cispa.de